

Sipeng Xie

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Systems research & engineering | Compiler/runtime · Parallel execution · Storage · AI infrastructure

Education

Beihang University, Beijing <i>Ph.D. in Cybersecurity, advised by Prof. Qianhong Wu</i>	2022 – Jun. 2027 (Expected)
Beihang University, Beijing <i>Graduate study in Cyberspace Security</i>	2021 – 2022
Beihang University, Beijing <i>B.E. in Information Security; second major in Mathematics. GPA 3.7625/4.0; Mathematics 95.33/100</i>	Sep. 2017 – Jul. 2021
Chengdu No. 7 High School (Linyin Campus) <i>NOIP Second Prize, 2014 and 2015</i>	Sep. 2014 – Jun. 2017

Publications

- **Sipeng Xie**, Qianhong Wu, et al. *SkelOT: Reusing AOT Compilation Across EVM Contract Families*. **EuroSys 2027 (accepted)**.
- **Sipeng Xie**, Qianhong Wu, et al. *MHOT: Height-Optimized Authenticated Data Structure for Blockchain State Commitment*. **USENIX Security 2026**.
- **Sipeng Xie**, Qianhong Wu, et al. *The Proxy Knows Too Much: Sealing LLM API Routers with Attested TEEs*. **Under review; arXiv 2026**.
- Wenkuan Xiao, Qianhong Wu, Wenbo Wu, **Sipeng Xie**, et al. *AccountCatcher: Anomaly Blockchain Account Detection Based on Hybrid Graph-Based Model*. **ProvSec 2025**.
- Tianxu Han, Qianhong Wu, Kun Wang, Yingmiao Zhang, **Sipeng Xie**. *Proof of Inaction for Outsourced Data with Exhaustive Audit*. **Computers & Electrical Engineering 2024**.
- Tianxu Han, Qianhong Wu, **Sipeng Xie**, et al. *Iterative Proof of Retrievability for Periodic Incremental Data via Optimistic Tags Generation*. **CSCloud 2024**.
- Kun Wang, Qianhong Wu, Tianxu Han, **Sipeng Xie**, et al. *PDTS: Practical Data Trading Scheme in Distributed Environments*. **ProvSec 2024**.
- Tianxu Han, Jian Mao, **Sipeng Xie**, et al. *VM-Studio: A Universal Crosschain Smart Contract Verification and Execution Scheme*. **Security and Communication Networks 2023**.

Engineering Experience

paradigmxyz/revmc Contributed five merged upstream PRs addressing LLVM undefined behavior, aliasing, resume-state correctness, and JIT performance. Replaced LLVM i256 DIV/MOD with ruint builtins (PR #88); PR benchmarks show Curve StableSwap JIT 109 → 36 μs and Uniswap V2 30 → 23 μs per transaction.	Rust / LLVM · 2026
Blockchain Execution Infrastructure (confidential) Supported early technical architecture and diligence, then contributed to about two years of execution-layer R&D. The project publicly announced an \$11M pre-seed round in 2025, co-led by Founders Fund and Pantera Capital.	2024 – 2026
Ant Group · Ant Star Internship <i>Ant Digital Technologies, Digital Technology Division; long-term internship offer.</i>	2027 · Incoming

AI and Developer Infrastructure

Pi Agent Runtime Learning Journal Built a TypeScript site with 18 source-driven lessons and Mermaid diagrams on agent loops, tool runtimes, context management, and session persistence.	Open source · 2026
Laboratory AI Router Operate shared multi-model API infrastructure for research and development; handle availability and incident response.	Beihang University · Ongoing

Technical Skills

Languages & tools: Rust, Python; working knowledge of C/C++ and TypeScript; LLVM, Linux, Git, code review.
Systems: Compiler/runtime optimization, parallel execution, authenticated storage, benchmarking.